

Computer Vision Engineer · AI/ML Developer

PROFESSIONAL SUMMARY

- Built a production footfall counting system through multiple iterations (v3–v10), evolving tracking from DeepSORT to BotSORT with camera motion compensation for stable counts on moving/vibrating cameras
- Designed dual-zone crossing logic with foot-centroid counting and occlusion-aware appearance freezing to prevent duplicate/missed counts during crowding
- Implemented frozen-frame detection in pre-buffers and async clip encoding via FFmpeg pipe, plus multi-camera parallelism using per-camera ORT sessions with ORT_ENABLE_BASIC for DirectML compatibility
- Built a companion Django REST backend (footfallai) with hour-range filtering across all endpoints for reporting and analytics

PROJECTS

Helmet Detection System — *YOLOv8 · OpenCV · Django · PostgreSQL · Streamlit*

- Built a real-time helmet/no-helmet classification system using YOLOv8 with live video frame-by-frame analysis via OpenCV
- Developed Django REST APIs for backend integration and stored detection results in PostgreSQL
- Designed an interactive Streamlit monitoring UI; improved detection accuracy through dataset preprocessing and iterative model tuning

EDUCATION

BE Computer Science and Engineering · Kingston Engineering College 2021 – 2025

KEY STRENGTHS

- Real-time multi-camera CV pipeline engineering
- Production debugging & root cause analysis
- End-to-end AI system development & deployment
- Multi-threaded GPU inference optimization
- Django REST API design & database integration
- Continuous learning & rapid iteration